



8493 - Spatially resolving obscured AGN candidates in galaxy mergers with JWST photometry

Cycle: 4, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

Galaxy mergers may play a critical role in the co-evolution of galaxies and their central supermassive black holes (SMBHs) by building stellar mass and triggering accretion onto SMBHs, visible as active galactic nuclei (AGN). The most rapid SMBH growth is predicted at late merger stages (<10 kpc galaxy separations) and under heavy obscuration. Currently, studies testing these predictions have relied on mid-infrared (MIR) photometric colors to identify heavily obscured AGN candidates in galaxy mergers. However, the angular resolution of those facilities (2-6") can not resolve the positions of AGN within late stage galaxy mergers for the vast majority of AGN. Therefore, these selections are unable to measure the merger stage or control for the stellar bulge mass of the AGN to isolate the merger dependence. To overcome these limitations, we propose an archival project to use JWST photometry, combined with the available HST photometry, to spatially resolve the positions of obscured and unobscured AGN candidates in galaxies from the COSMOS-Web treasury field based on sub-galactic modeling of spectral energy distributions. Through this project we will trace how SMBH growth evolves down to late merger stages for the most heavily obscured AGN and the dependence on merger mass ratio down to extreme values. The results produced through this project will also aid future development of spatially-resolved color-based AGN candidate selection from JWST photometry, provide constraints on the dominant growth routes of obscured SMBHs, and provide predictions of binary SMBH formation rates and expected gravitational wave signatures detectable by pulsar timing arrays and LISA.

OBSERVING DESCRIPTION